Diffusion Geometry for Phase Space Estimation

Topological and Differential Operators for Finding L

Notation & Geometric Foundations

Time-Delay Embedding:

$$X_t = [x_t, x_{t+\tau}, \ldots, x_{t+(L-1)\tau}] \in \mathbb{R}^L$$

Phase Space Velocity:

$$v_t = \frac{X_{t+1} - X_t}{||X_{t+1} - X_t||}$$

Coifman Normalisation ($\alpha = 1$):

$$K_{ij} = \exp\left(-\frac{||X_i - X_j||^2}{\sigma_i \sigma_j}\right)$$

$$q_i = \sum_j K_{ij}$$

$$P_{ij} = \frac{K_{ij}/(q_i q_j)}{\sum_k K_{ik}/(q_i q_k)}$$

1. Dirichlet Energy of Time

Chronological Node Function: Let t_i be the integer timestamp of the embedded state X_i. The normalised temporal function on the graph is:

$$f(X_i) = \frac{t_i}{N}$$

Discrete Dirichlet Energy:

$$\mathcal{E}_{\text{time}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j \in \mathcal{N}(i)} P_{ij} \left(\frac{t_j - t_i}{N} \right)^2$$

Mechanism: Evaluates macroscopic contextual resolution.
Objective: Minimise to penalise chronological looping and enforce global cyclic stretching.

2. Flow Roughness (Connection Laplacian)

Bochner / Connection Laplacian: $\nabla^* \nabla$

Intrinsic Alignment:

$$O_{ij} \in \mathrm{SO}(m) \quad \text{(Parallel Transport Matrix)}$$

Energy Functional:

$$\mathcal{E}_{\text{rough}} = \frac{1}{2N} \sum_{i=1}^{N} \sum_{j \in \mathcal{N}(i)} P_{ij} ||v_i - O_{ij} v_j||^2$$

Mechanism: Measures intrinsic vector field deviation.
Objective: Minimise to resolve local topological intersections and guarantee phase space unfolding.

3. Local Predictability Error

Diffusion-Weighted Future State:

$$\hat{v}_i = \frac{\sum_j P_{ij} v_j}{||\sum_j P_{ij} v_j||}$$

Prediction Error (Cosine Distance):

$$\mathcal{E}_{\text{pred}} = \frac{1}{N} \sum_{i=1}^{N} (1 - \langle v_i, \hat{v}_i \rangle)$$

Mechanism: Evaluates dynamical determinism (S-Map constraint).
Objective: Minimise to detect and penalise high-dimensional starvation.

4. Diffusion Graph Entropy

Shannon / Von Neumann Entropy on Transition Weights:

$$\mathcal{H}(P) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j\in\mathcal{N}(i)} P_{ij}\log P_{ij}$$

Mechanism: Quantifies structural uniformity of the phase space.
Objective: Maximise to prevent random walks from collapsing into artificial geometric hubs caused by false nearest neighbours.

5. Exact Energy (Hodge Divergence)

Discrete 1-Form (Edge Flow):

$$W_{ij} = \left\langle \frac{v_i + v_j}{2}, X_j - X_i \right\rangle_{\mathbb{R}^L}$$

Codifferential (Net Flux):

$$(\delta W)_i = \sum_{j \in \mathcal{N}(i)} P_{ij} W_{ij}$$

Exact Energy:

$$\mathcal{E}_{\text{exact}} = \frac{1}{N} \sum_{i=1}^{N} (\delta W)_i^2$$

Objective: Minimise spatial sources/sinks representing severe manifold crushing.

6. Harmonic Energy (Topological Cycles)

Poisson Equation (Exact Potential f):

$$(I - P)f = \delta W$$

Residual / Divergence-Free Flow:

$$R_{ij} = W_{ij} - (f_j - f_i)$$

Harmonic Proxy Norm:

$$\mathcal{E}_{\text{harmonic}} = \frac{1}{N} \sum_{i=1}^{N} \sum_{j \in \mathcal{N}(i)} P_{ij} R_{ij}^2$$

Objective: Maximise to isolate macroscopic 1D topological loops.

7. Spectral Gap (Laplacian Eigenvalues)

Random Walk Normalised Laplacian:

$$\Delta = I - P$$

Eigenvalue Spectrum:

$$0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$$

Spectral Gap:

$$\mathcal{S}_{\text{gap}} = \lambda_2 - \lambda_1$$

Mechanism: Tracks graph connectedness ($\lambda_1 > 0$) and symmetry.
Signature: $\mathcal{S}_{\text{gap}} \to 0$ signifies perfect topological loop.

8. Local Holonomy

Parallel Transport across 3-Cycles (Triangles):

$$H_{ijk} = O_{ij} O_{jk} O_{ki}$$

Holonomy Error:

$$\mathcal{E}_{\text{hol}} = \frac{1}{|\mathcal{T}|} \sum_{(i,j,k) \in \mathcal{T}} ||H_{ijk} - I_m||^2$$

Mechanism: Measures intrinsic manifold curvature.

9. Diffusion Distance Correlation

Ambient vs. Intrinsic Geometry:

$$A_{ij} = ||X_i - X_j||_{\mathbb{R}^L}$$
$$D_{ij} = ||P_{i,*} - P_{j,*}||_2$$

Where $P_{i,*}$ is the i-th row vector of the Markov transition matrix.

Correlation Metric:

$$\mathcal{C}_{\text{diff}} = \text{Corr}(A_{ij}, D_{ij})$$

Mechanism: Detects fundamental contradictions between Euclidean span and Markov transition time.